

Long-Term Vegetation Diversity Trends in North America: Evidence from Fossil Pollen

Thomas Gelgon

Exchange student Bachelor 3, Plymouth School of Geography

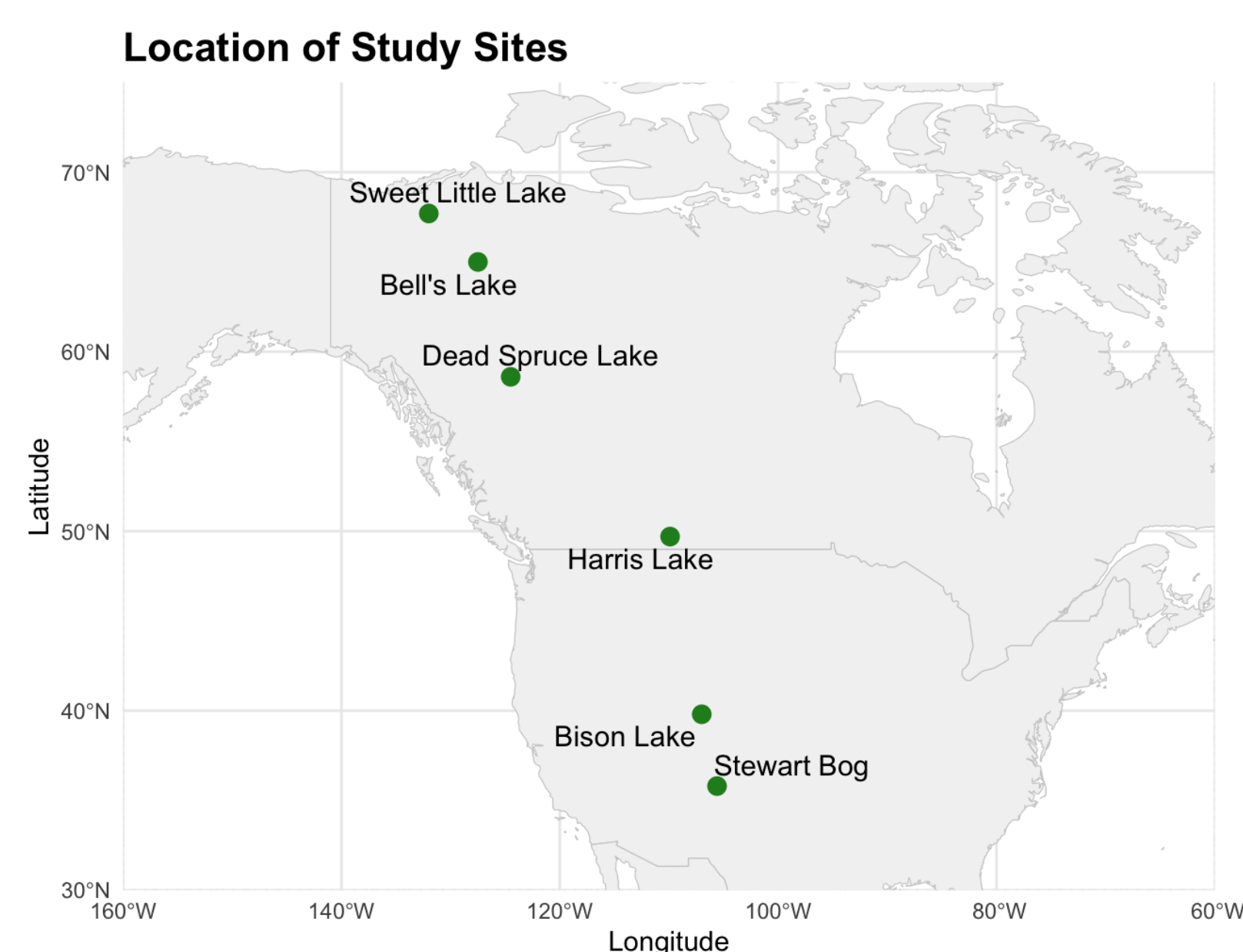
Introduction

Biodiversity is one of the major components of ecosystems resilience, stability and response to environmental change. However, scientist and decision makers usually rely on short-term data. Long-term data allows to comprehend how broader patterns and past biodiversity changes led to the current situation. Paleocological data such as fossil pollen records gives us insight into vegetation diversity changes spanning across centuries to millennia in response to climate change, local disturbances and human impact (Routson & al, 2021). In this research, we analyzed Simpson's diversity index from six North American lakes pollen records on a timespan of 11 000 years BP across a wide latitude gradient. We will explore Holocene broad patterns of diversity both regionally and for each site, timing of diversity maxima and differences in diversity values and variability among the sites.

Data and Method

- The data is fossil pollen counts
- 6 sites in North America: Sweet Little Lake, Bell's Lake, Dead Spruce Lake, Harris Lake, Bison Lake, Stewart Bog. Latitude range: 67.7°N to 35.8°N
- Temporal range covering the Holocene (11,000 years BP)
- The data underwent an analysis with R, pollen counts were converted to percentage and I calculated Simpson's diversity index through time for each site.

Figure 1: location of the 6 sites



- We then performed multiple compared analysis: diversity trends, time of peak diversity, diversity variability, diversity-latitude relations. These tests were conducted to evaluate the importance of local, latitudinal and climatic factors.

Patterns of diversity over the Holocene

The data analysis reveals a sharp decline in diversity in the two northern sites from 0.7 in 11000 BP to 0.22 in 9800 BP for Sweet Little Lake and 0.38 in 8800 BP for Bell's Lake, followed by a steep increase, a steep decrease and another increase (fig. 2). Around 5000 BP, both sites' diversity index (fluctuating between 0.65 and 0.75, up to 0.8 for Bell's Lake today) have exceeded their initial values and are closer to the other sites. These fluctuations reflects postglacial biodiversity adaptation and taxa changes (Betula and Shepherdia being replaced by Pinea and Alnus, as shown in fig. 3-1 & 3-2), as postglacial warming facilitated the northward expansion of forests (Dawson & al., 2024). This leads to biodiversity maturity and stabilization around 5000 BP. Sweet Little Lake being further North, it takes longer to achieve this transition.

The other sites show high diversity (0.76 - 0.86) at the beginning of the Holocene which is similar to their current diversity index. This reveals that the main driver of this sites are regional and local fluctuations instead of broader climatic patterns. However, Stewart Bog and Harris Lake have more intense fluctuations from 3500 BP which while Dead Spruce Lake and Bison Lake are very stable.

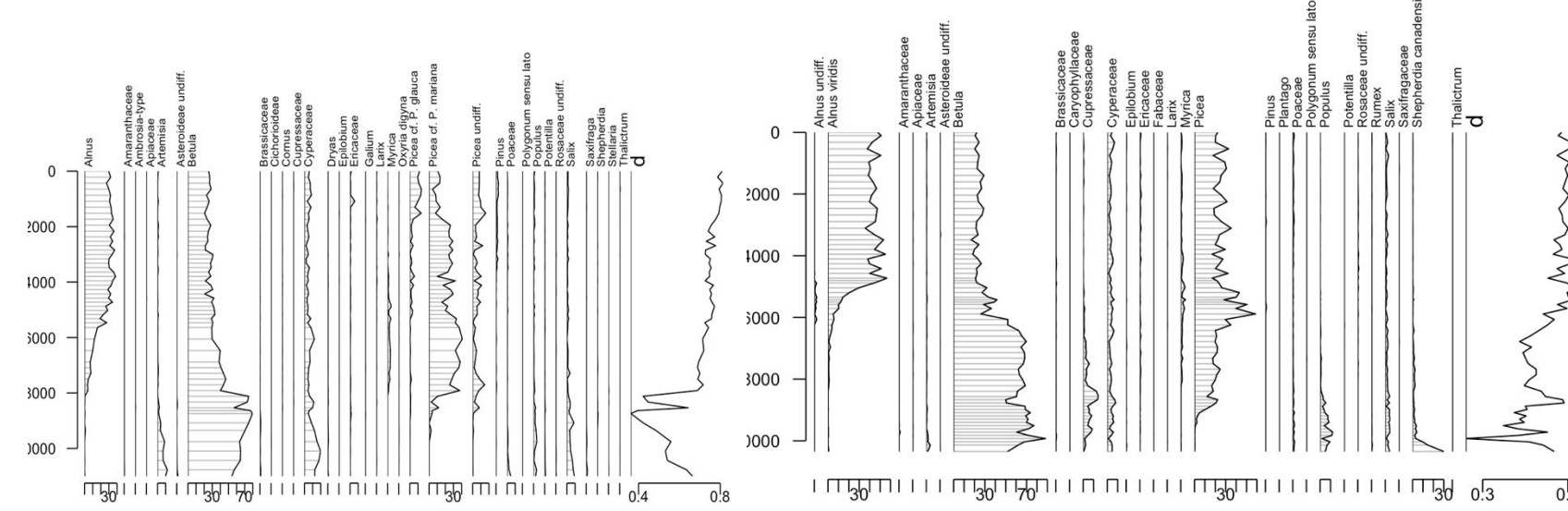


Figure 3-1: Taxa changes at Bell's Lake

Figure 3-2: Taxa changes at Sweet Little Lake

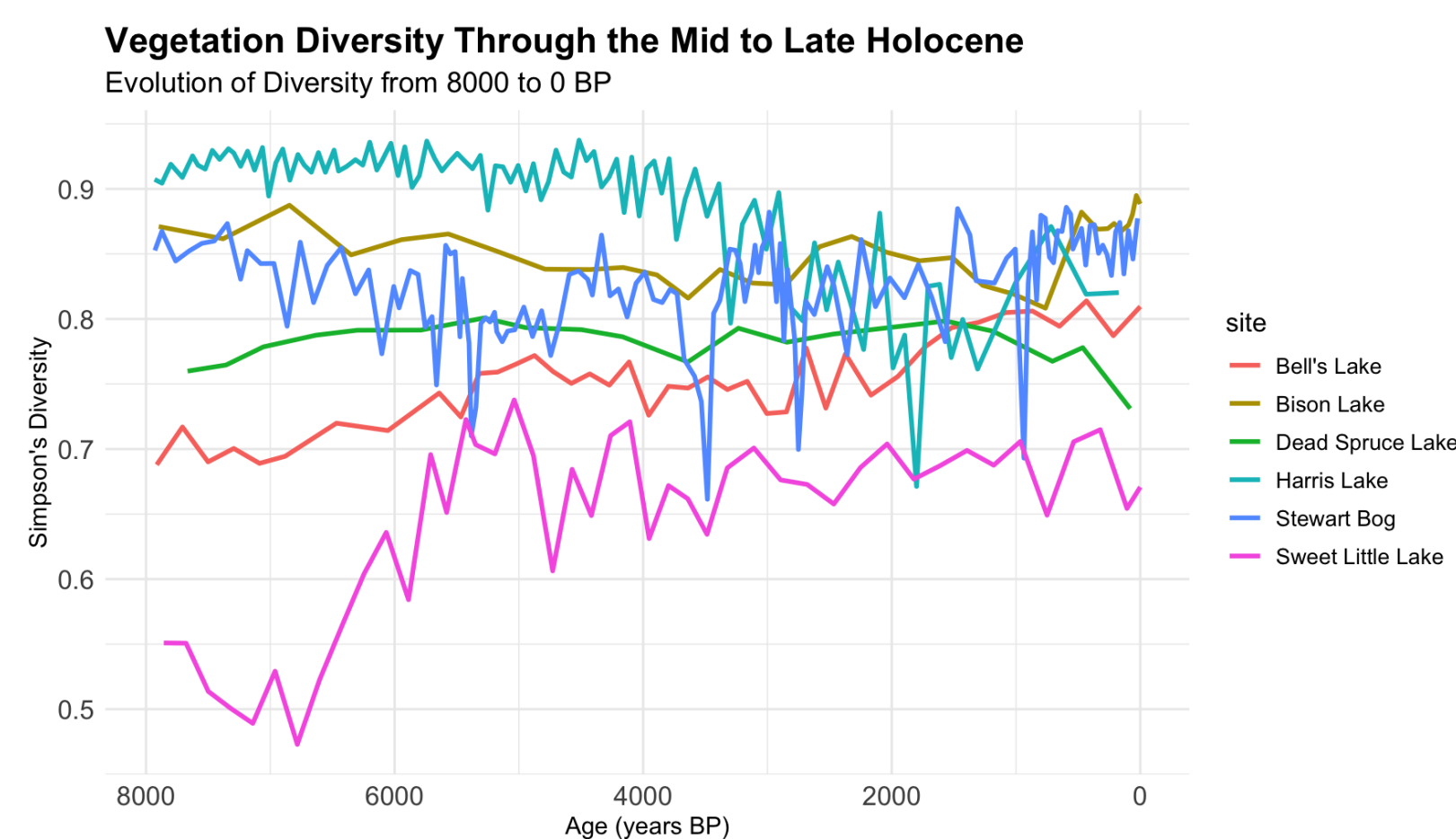


Figure 2: evolution of diversity by site

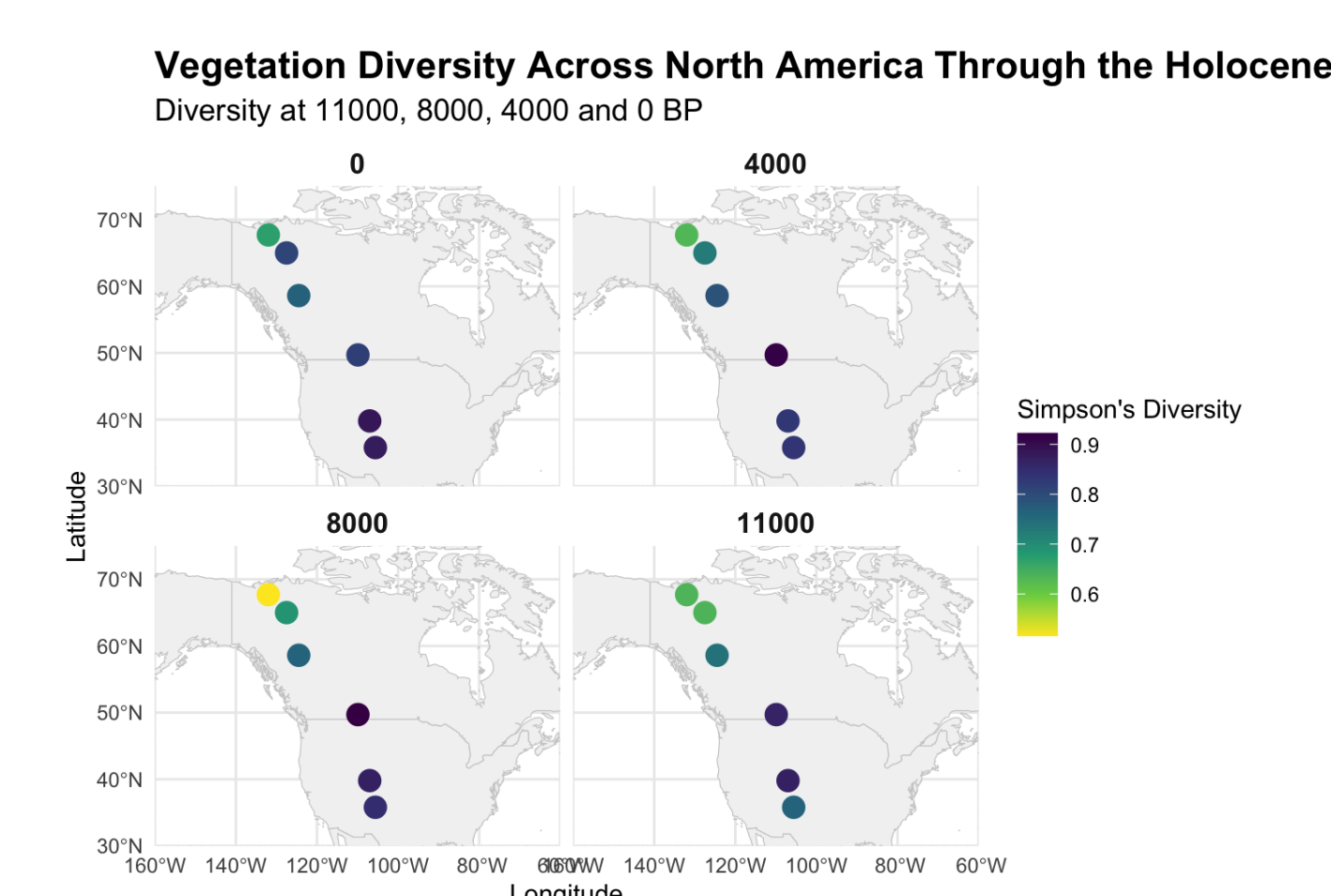


Figure 4: map of evolution of diversity by site. We notice that the two northern sites have a very different pattern from the other ones.

Latitudinal Gradient in Vegetation Diversity

Lower diversity at higher latitudes?

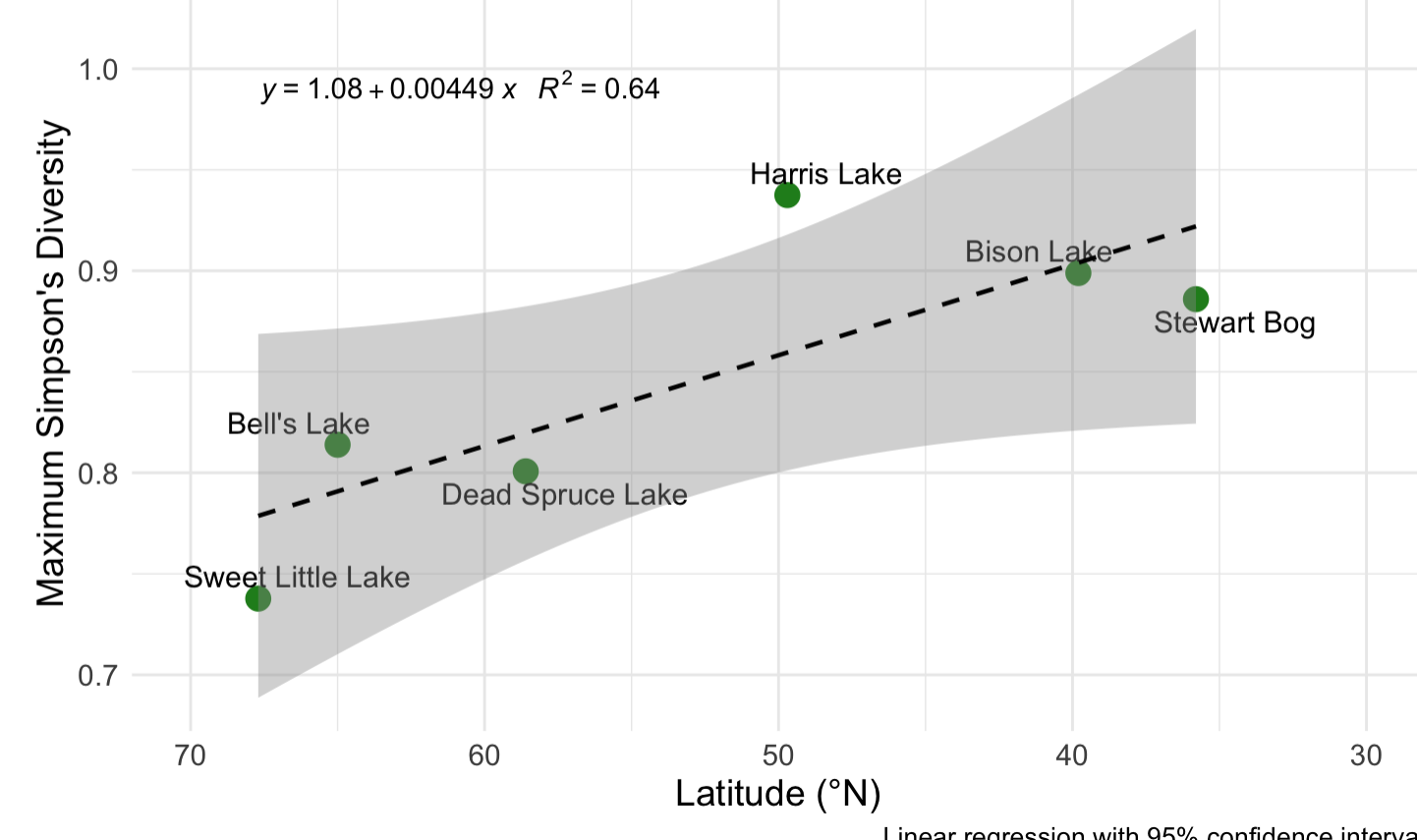
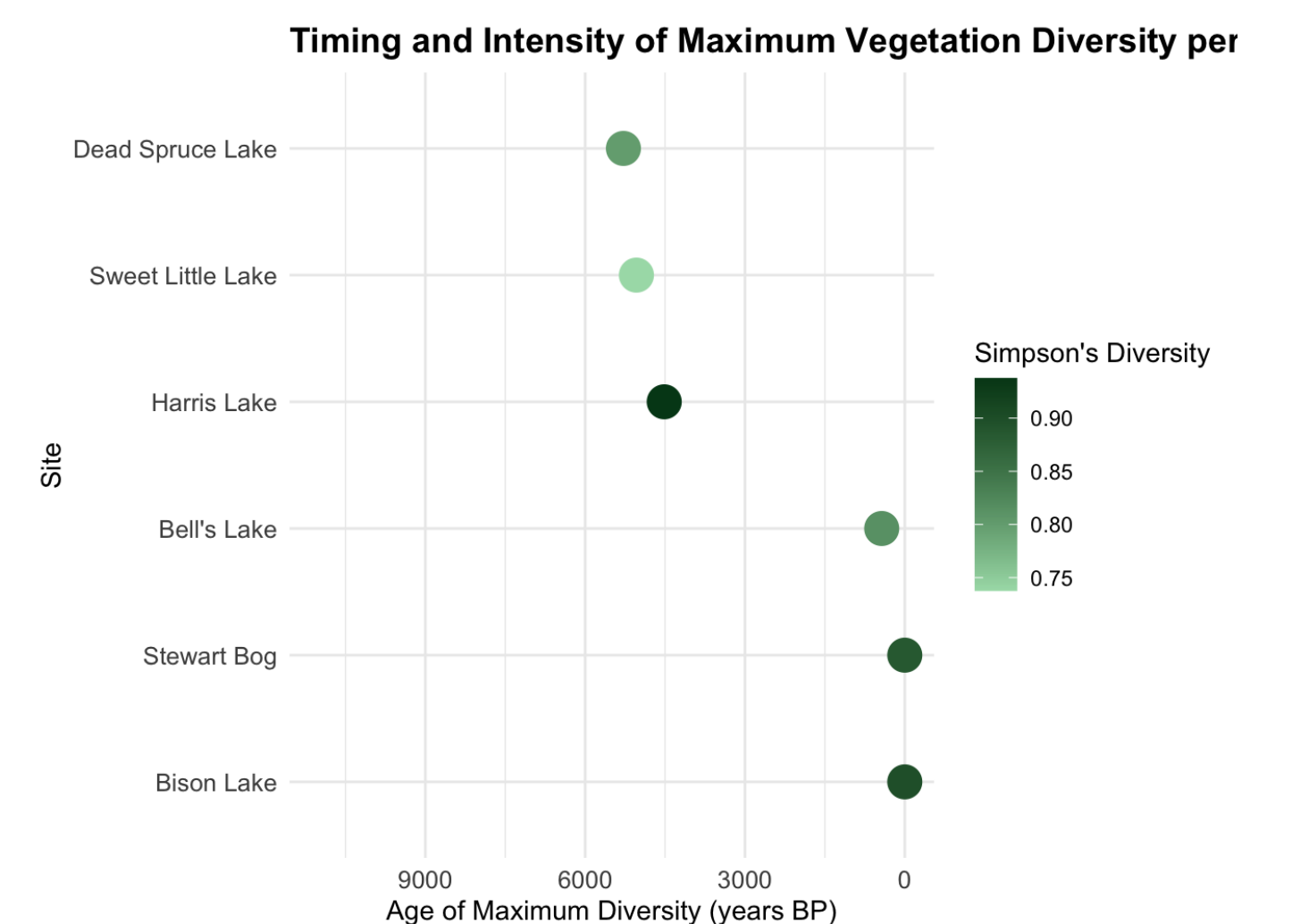


Figure 5: we see a strong correlation between latitude and diversity, with minor variations due to local condition. Harris Lake have the most central location which allows the broadest taxa to colonize this area.

Peak diversity among different sites

Knowing an ecosystem's peak diversity gives us a precious insight into its evolution across the Holocene. We can classify our sites in two groups. Group 1 (Dead Spruce Lake with 0.801, Sweet Little Lake with 0.738, Harris Lake with 0.937) peaks during mid-holocene climate optimum (mild, stable conditions), respectively, 5282, 5038 and 4516 BP. This suggest an early ecological stabilization (See fig 6).

Figure 6 clearly show two distinct trends among the sites, but peak diveristy can be various among those groups.



On the other hand, group 2 (Bell's Lake with 0.814, Bison Lake with 0.899, Stewart Bog with 0.886) reach their peak much later, respectively, 433, 31 and 29 BP. This reflects continued ecological diversification as fig 3-1, 7-1 and 7-2 show. Bell's Lake, despite being very northern, peaks very late. This could be explained by hydrological conditions (it is close to Mackenzie river) or competition between species (other Picea seem to have replaced Picea cf P. Mariana as we see on fig 7-1) or neoglaciation (uneven across North America) which would have led to colder temperatures (Briner & Porter, 2018).

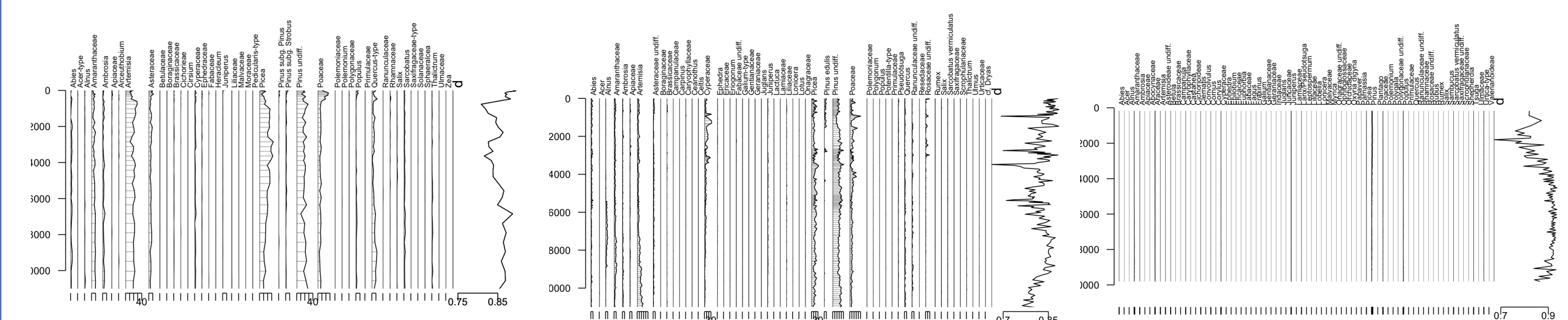


Figure7-1: Taxa changes at Bison Lake

Figure7-2: Taxa changes at Stewart Bog

Figure 7-3 shows the huge taxa diversity at Harris Lake

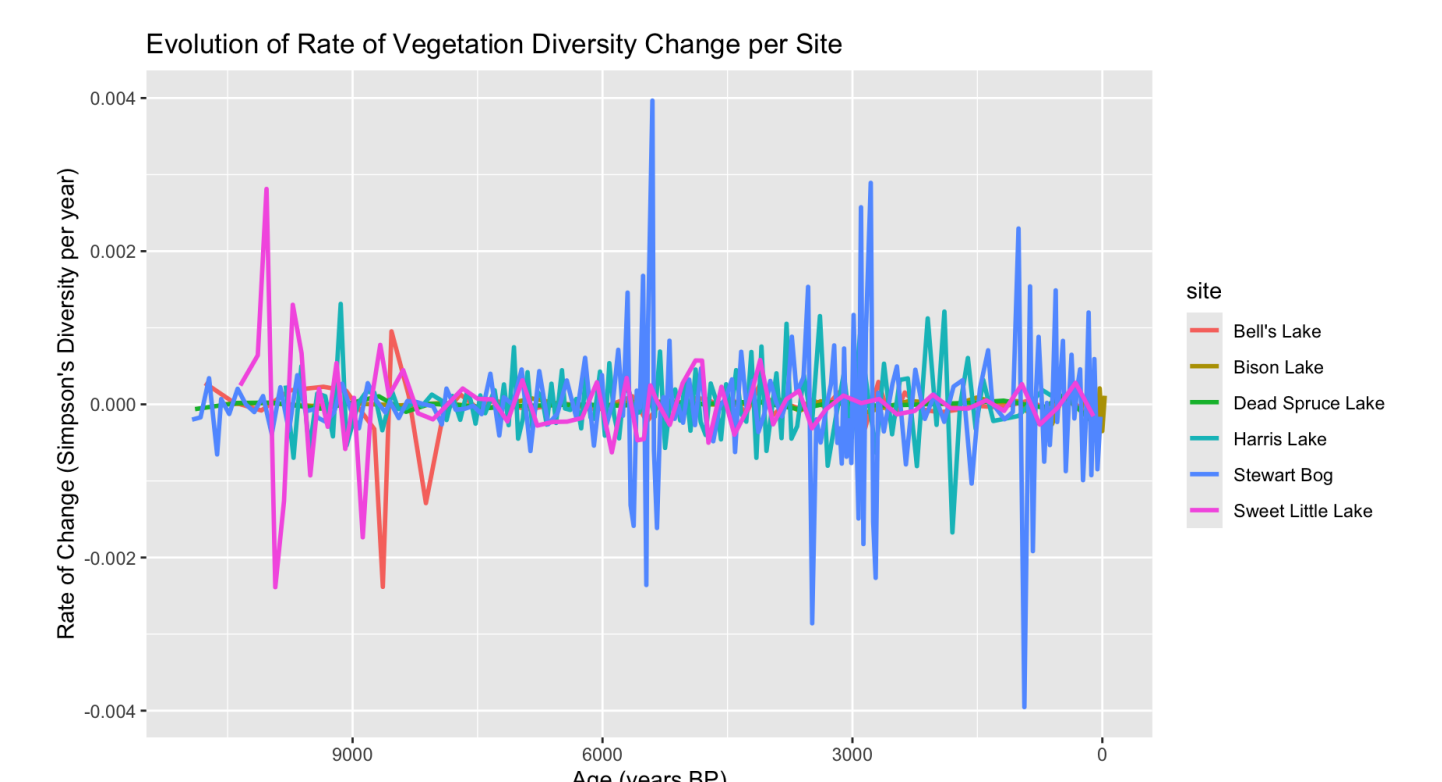
Diversity values

After taking a closer look at the different sites' diversity values (initial value, peak diversity, variability), we will discuss again on the possible conditions of these differences. As we said previously, early Holocene values are very contrasted. A first group (Bell's and Sweet Little Lakes) start lower (0.62) and experience a quick drop (see fig. 2). The other sites range between 0.72 and 0.94 and with relatives stables values will keep this stability until 3500 BP. From this time, Harris Lake (which had the highest diversity index until then, mostly higher than 0.9) will slightly decline to 0.82 nowadays (with a drop to 0.67). Stewart Bog experiences much more fluctuations and together with Bison Lake (the two southernmost sites) their diversity will rise during the 700 last years, maybe as a consequence of european colonization which brought new species to colonize the areas. During the 5500 last years, Bell's Lake diversity will increase from 0.75 to 0.81 while Sweet little Lake diversity's will decrease slightly from 0.73 to 0.67. This is shown on fig. 3-1 and 3-2 where we see new species rising at Bell's Lake from the last millennia while it is not the case at Sweet Little Lake.

Maximum diversity values offer us a gradient with a strong correlation between mid to mid-high latitudes and high diversity: Harris (0.937), Bison (0.899), Stewart (0.886), Bell's (0.814), Dead Spruce (0.801), Sweet (0.738). Harris Lake, which is at the transition zone (49.7°N) between boreal and temperate forests zones (see fig. 7-3), has the highest diversity.

If we look at the variability index (fig. 8), we clearly see the changes of Bell's Lake and Sweet Little Lake. The even higher shifts of Stewart Bog indicate frequent but short-term changes, probably caused by natural hazards such as fires.

Figure 8:variability of diversity



Conclusions

The comparison of the six sites reveals that there is no single unified pattern of diversity. Biodiversity trajectories varies site by site, depending on latitude, postglacial history and local specificities. While northern sites experience low and unstable early diversity caused by deglaciation and its delayed effects, mid and southern sites show early high diversity and their assembly is more similar to what it was ten millennia ago. We can thus establish a broad latitudinal gradient with southern sites exhibiting higher diversity as Giesecke & al. Has also shown in their work on Europe (Giesecke & al., 2019). However, there are also differences in peak diversity not strictly correlated with latitude, which proves that latitude is not the only factor underlying diversity. As a matter of fact, Dead Spruce Lake, while being located at relatively high latitude (58°N), is ecologically very stable, hence showing the role of local conditions. The same goes for Bell's Lake whose pattern is different from other northern sites. As other studies show, Holocene biodiversity patterns are context-dependent and multi-causal (Birks & al., 2016). To conclude, we can acknowledge that this piece of research shows the importance of paleoecological data to help us to understand the complexity of biodiversity and climate science.

References

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